Harvard Griffin GSAS — Mathematics (PhD)

Statement of Purpose: Discrete Mathematics, Spectral Graph Theory, and Stochastic Optimization

Candidate:	Sham Satish Thakare (PRN: 72159719L, Seat: B191102032)
Target Program:	PhD in Mathematics, Harvard Kenneth C. Griffin GSAS
Application Deadline:	December 15, 2026 (5:00 PM EST)
Undergraduate Degree:	B.E. in AI & DS, Savitribai Phule Pune University (APCOER Pune, 2020–2024)
Academic Distinctions:	Gold Medal in Mathematics (2024), 8.70 CGPA (First Class with Distinction), 9.53 Final SGPA (Rank 1)
Target Faculty:	Prof. Horng-Tzer Yau, Prof. Noam Elkies, Prof. Lauren Williams

Statement of Purpose

From the structural symmetry of algebraic graph spectra to the measure-theoretic convergence of stochastic sequences, my intellectual journey has been driven by a conviction that complex computational phenomena are governed by rigorous mathematical principles. Awarded the university Gold Medal in Mathematics (2024) across my graduating cohort at Savitribai Phule Pune University, I have spent the past two years formalizing mathematical structures that underlie distributed systems and stochastic optimization algorithms. At the Harvard Griffin Graduate School of Arts and Sciences, I seek to pursue doctoral research in discrete mathematics, spectral graph theory, and mathematical optimization, investigating the analytical invariants that guarantee convergence, stability, and topological integrity in high-dimensional stochastic processes.

My foundational coursework in mathematics—spanning Discrete Mathematics, Real and Complex Analysis, Linear Algebra, and Numerical Methods—instilled in me a commitment to axiomatic precision. In my recent research on autoregressive token generation sequences, I approached the ubiquitous empirical problem of reasoning calibration not as a heuristic engineering challenge, but through the lens of analytical probability theory and information theory. By examining the conditional Shannon entropy $H(Y_t \mid Y_{<t}, X)$ along reasoning trajectories, I proved that cumulative sequence entropy exhibits strong positive collinearity with trajectory path length ($r = +0.486$, $p < 10^{-6}$). This analytical finding proved that prevailing heuristic uncertainty weighting in reinforcement learning serves not as an epistemic uncertainty measure, but as an implicit length penalty. Formalizing this distinction required establishing measure-theoretic bounds on conditional probability measures across discrete sequence spaces, confirming that stochastic calibration requires invariant geometric metrics rather than scalar token entropies.

Simultaneously, my work in distributed state-machine consensus required deriving topological proofs for dynamic quorum intersections. In asynchronous networks governed by bounded drift Delta and arbitrary packet delays, maintaining linearizability under Byzantine or network partition regimes reduces to establishing an invariant non-empty intersection between any two dynamically formed quorums: $Q_i \cap Q_j \neq \emptyset$. By formulating dynamic quorum allocation as a hypergraph coloring problem over time-varying communication graphs, I derived an analytical proof demonstrating that even under asymmetric link degradation, dynamic quorum re-weighting strictly preserves the majority intersection invariant without violating safety invariants. This bridge

between combinatorial graph theory and distributed linearizability cemented my ambition to pursue foundational mathematical research.

Harvard's Department of Mathematics represents the ideal intellectual environment for my doctoral training. I am particularly drawn to the groundbreaking work of Prof. Horng-Tzer Yau at the interface of probability theory, random matrix theory, and mathematical physics. I am eager to explore how universality classes of random matrices and spectral statistics can model the convergence dynamics of high-dimensional gradient systems and non-convex optimization landscapes. Furthermore, the pioneering contributions of Prof. Noam Elkies in discrete mathematics, algebraic combinatorics, and computational number theory deeply inspire my desire to explore extremal graph properties and lattices in discrete optimization. Additionally, I would be privileged to investigate combinatorial and geometric structures under Prof. Lauren Williams, particularly positive Grassmannians and cluster algebras, exploring their connections to combinatorial optimization.

With a proven record of mathematical rigor, university honors, and a sustained drive for formal proof, I am prepared to contribute meaningfully to the scholarly community of Harvard's Department of Mathematics.